Tub sonlar, tub ko'paytuvchilarga ajratish va daraja

1-test (oson) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

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1. **Qaysi son 6 ning karralisi?**

A) 87 **B)** 78 **C)** 92 **D)** 64

2. **Qaysi son tub son?**

A) 51 **B)** 39 **C)** 57 **D)** 67

3. **Qaysi son murakkab son?**

A) 77 **B)** 43 **C)** 29 **D)** 2

4. **1 soni haqida qaysi gap to'g'ri?**

A) 1 — tub son
B) 1 — eng kichik tub son
C) 1 tub son ham, murakkab son ham emas
D) 1 — murakkab son

5. **Eng kichik tub son qaysi?**

A) 2 **B)** 1 **C)** 3 **D)** 0

6. **50 ning barcha bo'luvchilari qaysi qatorda yozilgan?**

A) 2, 5, 10, 25
B) 1, 2, 5, 10, 25
C) 1, 2, 5, 10, 25, 50
D) 50, 100, 150, 200

7. $3 \cdot 3 \cdot 3 \cdot 3 \cdot 3$ **ko'paytmani daraja shaklida yozing.**

A) 3^4 **B)** 3^5 **C)** 5^3 **D)** 15

8. 8^5 **darajada asos va ko'rsatkich qaysi?**

A) asosi 8, ko'rsatkichi 5
B) asosi 40, ko'rsatkichi 5
C) asosi 8, ko'rsatkichi 40
D) asosi 5, ko'rsatkichi 8

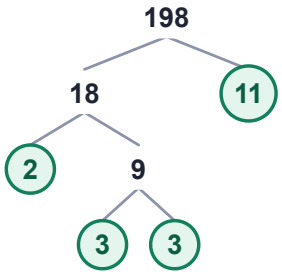
9. **11 ning kvadrati nechaga teng?**

A) 22 **B)** 13 **C)** 1 331 **D)** 121

10. **5 ning kubi nechaga teng?**

A) 25 **B)** 15 **C)** 125 **D)** 625

11. **Rasmda 198 ning ko'paytuvchilar daraxti berilgan. 198 ning tub ko'paytuvchilarga yoyilmasi qaysi?**



A) $2 \cdot 3 \cdot 11$ **B)** $18 \cdot 11$

C) $2 \cdot 3^2 \cdot 11$ **D)** $2 \cdot 9 \cdot 11$

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12. **Qaysi son 5 ga qoldiqsiz bo'linadi?**

A) 5 321 **B)** 2 315 **C)** 3 452 **D)** 1 234

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13. **Tub sonning nechta bo'luvchisi bor?**

A) ikkitadan ko'p **B)** 3 ta

C) 1 ta **D)** 2 ta

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14. **Qaysi yozuv 36 ning tub ko'paytuvchilarga yoyilmasi?**

A) $2 \cdot 2 \cdot 9$ **B)** $2 \cdot 2 \cdot 3 \cdot 3$

C) $6 \cdot 6$ **D)** $4 \cdot 9$

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15. **Rasmda 390 qisqa bo'lish usulida ajratilgan. 390 ning tub ko'paytuvchilari qaysilar?**

390	2
195	3
65	5
13	13
1	

A) 2, 3, 13 **B)** 1, 2, 3, 5, 13

C) 2, 3, 5, 13 **D)** 195, 65, 13

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16. **Qaysi son 9 ning karralisi emas?**

A) 63 **B)** 94 **C)** 27 **D)** 81

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17. **Son 14 ga qoldiqsiz bo'linadi. U yana qaysi songa albatta bo'linadi?**

A) 4 **B)** 3 **C)** 7 **D)** 28

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18. **Daraja ko'rsatkichi nimani bildiradi?**

A) asosga nechani qo'shish kerakligini

B) sonning nechta bo'luvchisi borligini

C) asos o'z-o'ziga necha marta ko'paytirilganini

D) asosni nechaga ko'paytirish kerakligini

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19. **Ko'paytuvchilar daraxtida qanday son chiqsa, u shox davom ettirilmaydi?**

A) toq son **B)** 10 dan kichik son

C) juft son **D)** tub son

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20. 43 ta kitobni bir nechta javonga teng terish kerak: javonlar 1 tadan ko'p, har javonda ham 1 tadan ko'p kitob bo'lsin. Buni qilish mumkinmi?

A) Ha, 7 ta javonga

B) Ha, 3 ta javonga

C) Ha, 4 ta javonga

D) Yo'q, chunki 43 — tub son

Javoblar

Tub sonlar, tub ko'paytuvchilarga ajratish va daraja · 1-test (oson)

№	Javob	Qiymati
1	**B**	78
2	**D**	67
3	**A**	77
4	**C**	1 tub son ham, murakkab son ham emas
5	**A**	2
6	**C**	1, 2, 5, 10, 25, 50
7	**B**	3^5
8	**A**	asosi 8, ko'rsatkichi 5
9	**D**	121
10	**C**	125

№	Javob	Qiymati
11	**C**	$2 \cdot 3^2 \cdot 11$
12	**B**	2 315
13	**D**	2 ta
14	**B**	$2 \cdot 2 \cdot 3 \cdot 3$
15	**C**	2, 3, 5, 13
16	**B**	94
17	**C**	7
18	**C**	asos o'z-o'ziga necha marta ko'paytirilganini
19	**D**	tub son
20	**D**	Yo'q, chunki 43 — tub son

Qisqa yechimlar

1. 6 ning karralisi 6 ga qoldiqsiz bo'linadi.
78 : 6 = 13, ya'ni 78 = 6 · 13.
Javob: **78**. (64, 87, 92 ni 6 ga bo'lsak, qoldiq qoladi.)

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2. 51 = 3 · 17, 57 = 3 · 19, 39 = 3 · 13 — ularning hammasi 3 ga bo'linadi.
67 esa 2, 3, 5, 7 ga bo'linmaydi, bo'luvchilari faqat 1 va 67.
Javob: **67**.

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3. 77 = 7 · 11, bo'luvchilari: 1, 7, 11, 77 — ikkitadan ko'p.
Javob: **77**. (2, 29, 43 — tub sonlar.)

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4. Tub sonning ikkita bo'luvchisi bor, 1 ning esa bittasi — o'zi.
Demak, **1 tub son ham, murakkab son ham emas**. Eng kichik tub son — 2.

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5. 0 va 1 tub son emas.
2 ning bo'luvchilari faqat 1 va 2 — u tub.
Javob: **2** (yagona juft tub son).

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6. 50 = 1 · 50 = 2 · 25 = 5 · 10.
Bo'luvchilar: **1, 2, 5, 10, 25, 50** — 1 va sonning o'zi ham kiradi.
50, 100, 150, 200 — 50 ning karralilari.

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7. 3 soni 5 marta ko'paytirilgan: asosi 3, ko'rsatkichi 5.
Javob: 3^5.

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8. Pastdagi katta son — asos, yuqoridagi kichik son — ko'rsatkich.
$8^5 = 8 \cdot 8 \cdot 8 \cdot 8 \cdot 8$: **asosi 8, ko'rsatkichi 5**.

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9. Kvadrat — 2-daraja: $11^2 = 11 \cdot 11$ = **121**.
22 = 11 · 2 — bu daraja emas; 1 331 — 11 ning kubi.

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10. Kub — 3-daraja: $5^3 = 5 \cdot 5 \cdot 5$.
5 · 5 = 25, 25 · 5 = **125**.
15 = 5 · 3 — bu xato: daraja ko'paytma emas.

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11. Daraxt uchidagi tub sonlar: 2, 3, 3, 11.
198 = 2 · 3 · 3 · 11 = $2 \cdot 3^2 \cdot 11$.
18 va 9 — murakkab, ular yoyilmaga kirmaydi.

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12. Oxirgi raqami 0 yoki 5 bo'lgan son 5 ga bo'linadi.
Faqat **2 315** 5 bilan tugaydi.

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13. Tub son faqat 1 ga va o'ziga bo'linadi.
Masalan, 13: bo'luvchilari 1 va 13 — **2 ta**.
Ikkitadan ko'p bo'luvchisi bor son — murakkab.

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14. Yoyilmada hamma ko'paytuvchi tub bo'lishi kerak.
4, 6, 9 — murakkab sonlar.
Javob: **2 · 2 · 3 · 3**.

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15. Tub ko'paytuvchilar — tik chiziqning o'ng tomonidagi sonlar: **2, 3, 5, 13**.
Chapdagi 195, 65, 13 — bo'linmalar; 1 tub son emas.
Tekshirish: 2 · 3 · 5 · 13 = 390.

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16. 27 = 9 · 3, 63 = 9 · 7, 81 = 9 · 9.
94 ni 9 ga bo'lsak, 4 qoladi.
Javob: **94**.

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17. 14 = 2 · 7. Son 14 ga bo'linsa, 14 ning tub bo'luvchilariga — 2 ga va 7 ga ham bo'linadi.
Javob: **7**.
Masalan, 14 son 4 ga ham, 28 ga ham, 3 ga ham bo'linmaydi.

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18. $4^3 = 4 \cdot 4 \cdot 4$: ko'rsatkich 3 — 4 soni 3 marta ko'paytirilgan.
Javob: **asos o'z-o'ziga necha marta ko'paytirilganini**.
4 · 3 = 12 emas, $4^3 = 64$.

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19. Tub sonni yana ajratib bo'lmaydi — shox to'xtaydi.
Javob: **tub son**.
9 — toq va 10 dan kichik, lekin 9 = 3 · 3, uni ajratamiz.

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20. 43 ning bo'luvchilari faqat 1 va 43 — u tub son.
Demak, 43 ni 1 dan katta ikki son ko'paytmasi qilib yozib bo'lmaydi.
Javob: **yo'q**.
